

(22.0 Points)

1) Suppose you have a perceptron with two inputs, weights [$w_1=0.4$, $w_2=-0.7$], and a bias of 0.2 and the activation function is ReLU.

Compute the output when the inputs are [$x_1=1$, $x_2=0$]. (2.0 points)

2) Suppose you have a binary classification task. Your model predicts a probability of 0.8 for the positive class, but the true label is 1.

Compute the binary cross-entropy loss for this prediction (please write the formula). (2.0 points)

3) Outline the steps involved in training a neural network. Briefly explain the role of each step (2.0 points).

4) Suppose you are training a deep neural network, but it fails to learn: the gradients become very small as you move backward through the network.

What is this phenomenon called? Suggest two solutions to address it (2.0 points).

5) You have two models for an image classification task:

- Model A: A deep fully connected network.
- Model B: A convolutional neural network (CNN) with fewer parameters.

Which model is likely to perform better and why? (2.0 points)

6) Imagine a convolutional layer with:

- Input size: 32×32 ,
- Filter size (kernel): 5×5 ,
- Stride: 1,

(a) What will be the size of the output feature map without padding? (1.0 points)

(b) What will be the size of the output feature map with padding of 2? (1.0 points)

(c) Apply a 2×2 Max Pooling and a 2×2 Average Pooling (with stride = 2) to the following 4×4 Convolutional output/feature map.

Determine the output dimensions and fill in the resulting matrix. (2.0 points)

4x4 convolutional output

12	5	0	6
25	8	18	12
12	14	85	4
17	32	12	50

2x2 max pooling

?x? pooling output

4x4 convolutional output

12	5	0	6
25	8	18	12
12	14	85	4
17	32	12	50

2x2 average pooling

?x? pooling output

7) You observe that your CNN has a high performance on the train set, but a low performance on the validation set. What is this phenomenon called? (0.5 points)

List three methods you can apply to address this. (1.5 points)

8) What is the receptive field of a neuron? How is it conceptually different between a Convolutional Neural Network (CNN) and a Graph Convolutional Network (GCN)? (2.0 points)

9-1) What is the main difference between a regular autoencoder and a variational autoencoder (VAE)?

9-2) How does this difference impact the use of VAEs for generative tasks? (2.0 points)

10) Explain the purpose of the KL divergence term in the Variational Autoencoder (VAE) loss function. What happens if we remove the KL divergence part? (2.0 points)
